

6

Fundamentals of Drives

- DC Drive consists of a DC motor, power electronic converter. i.e. rectifier (or) chopper.
- Speed sensing mechanism – tachometer and feed back circuit and intelligent device (Micro controller) The following DC motors are suitable for speed control applications

- (a) series motor
- (b) separately excited DC motor.

- In the speed control applications, load remains same i.e., output current is continuous and assumed to be constant.

$$E_a = \frac{Z\phi NP}{60A}$$

$$\omega_m = 2\pi n$$

$$E_a = Z\phi \left(\frac{\omega_m}{2\pi} \right) \left(\frac{P}{A} \right)$$

$$= \left(\frac{Z}{2\pi} \cdot \frac{P}{A} \right) \phi \omega_m$$

$$E_a = K_a \phi \omega_m$$

$$\text{where } K_a = \frac{Z}{2\pi} \left(\frac{P}{A} \right)$$

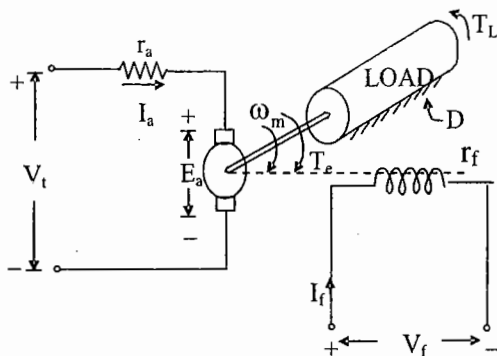
$$E_a \cdot I_a = T_e \omega_m$$

(Electrical Power) (mechanical power)

$$T_e = \frac{k_a \phi \omega_m I_a}{\omega_m}$$

$$T_e = k_a \phi I_a$$

Separately excited DC motor:



$$E_a = K_a \phi \omega_m$$

$$= (K_a \phi) \omega_m$$

$$E_a = K_m \omega_m$$

$$\text{where } K_m = K_a \phi \left[\frac{V-S}{\text{rad}} \right]$$

$$T_e = K_a \phi I_a$$

$$T_e = K_m \cdot I_a$$

$$\text{where } K_m = \left[\frac{\text{N.m}}{\text{A}} \right]$$

DC series motor :

$$E_a = K_a \phi \omega_m$$

$$\phi \propto I_a$$

$$\phi = c I_a$$

$$E_a = K_a c I_a \omega_m$$

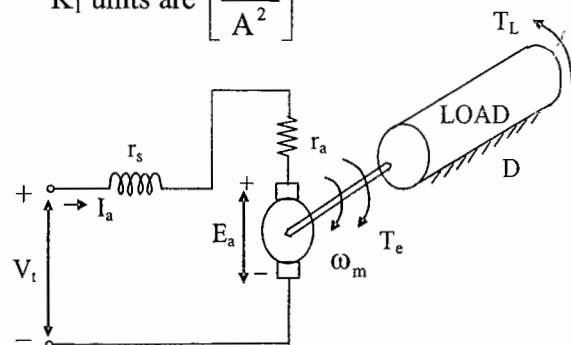
$$E_a = K_1 I_a \omega_m$$

$$T_e = K_a \phi I_a$$

$$= K_a c I_a \cdot I_a$$

$$T_e = K_1 I_a^2$$

$$K_1 \text{ units are } \left[\frac{\text{N.m}}{\text{A}^2} \right]$$





Workedout Examples

01. A d.c. series motor, fed from 400 V d.c. source through a chopper, has the following parameters:

$r_a = 0.05 \Omega$, $r_s = 0.07 \Omega$, $K = 5 \times 10^{-3}$ Nm/amp². The average armature current of 200 A is ripple free. For a chopper duty cycle of 50%, determine

- (a) input power from the source
(b) motor speed and motor torque

Sol: Given data,

$$V_s = 400V, I_a = 0.05\Omega, r_s = 0.07\Omega$$

$$K_t = 5 \times 10^{-3} \text{ Nm/A}^2, I_a = 200A$$

$$\alpha = 0.5,$$

$$V_0 = \alpha V_s$$

$$= 0.5 \times 400 \Rightarrow 200V$$

a) $V_t = 200V$

$$P = V_t I_a = 200 \times 200 = 40kW$$

b) $V_0 = V_t = E_a + I_a(R_a + R_s)$

$$= K_t I_a \omega_m + I_a(R_a + R_s)$$

$$200 = 5 \times 10^{-3} \times 200 \times \omega_m + 200(0.05 + 0.07)$$

$$\omega_m = 176$$

$$2\pi N = 176 \times 60$$

$$N = 1681 \text{ rpm}$$

Motor torque

$$T = K_t I_a^2$$

$$= 5 \times 10^{-3} \times (200)^2$$

$$= 200 \text{ N-m}$$

02. A dc chopper is used for regenerative braking of a separately excited dc motor. The dc supply voltage is 440V. The motor has $r_a = 0.3\Omega$, $K_m = 1.3$ V.s/rad. The average armature current during regenerative braking is kept constant at 250 A with negligible ripple. Find minimum and maximum permissible breaking speeds.

- (a) 969 r.p.m, 3783 r.p.m
(b) 969 r.p.m, 2681 r.p.m
(c) 550 r.p.m, 2681 r.p.m
(d) 550 r.p.m, 3783 r.p.m

Sol: Minimum breaking speed is

$$\omega_{\min} = \frac{I_a r_a}{K_m} = \frac{250 \times 0.3}{1.3} = 57.69 \text{ rad/sec}$$

$$\frac{2\pi N}{60} = 57.69$$

$$N = 550.89 \text{ r.p.m}$$

Maximum braking speed is

$$\omega_{\max} = \frac{V_s + I_a r_a}{K_m} = \frac{440 + (250 \times 0.3)}{1.3}$$

$$= 396.15 \text{ rad/sec}$$

$$= 3782.9 \text{ r.p.m}$$

03. An electric motor, developing a starting torque of 15 Nm, starts with a load torque of 7 Nm on its shaft. If the acceleration at start is 2 rad/sec², the moment of inertia of the systems must be (neglecting viscous and Coulomb friction)

(GATE - 05)

(a) 0.25 kgm²

(b) 0.25 Nm²

(c) 4 kg-m²

(d) 4 Nm²

Sol: $T_e = 15 \text{ Nm}$

$T_L = 7 \text{ Nm}$

$$\frac{dw}{dt} = 2 \text{ rad/sec}^2 \quad J = ?$$

$$\tau_e = J \frac{d\omega}{dt} + D\omega + T_L$$

$$15 = J(2) + 7 \Rightarrow J = 4 \text{ kg m}^2$$

04. A variable speed drive rated for 1500rpm, 40 Nm is reversing under no load. Figure shows the reversing torque and the speed during the transient. The moment of inertia of the drive is

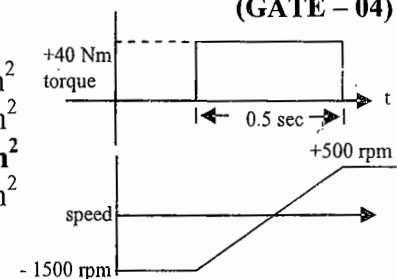
(GATE - 04)

(a) 0.048 kg-m²

(b) 0.064 kg-m²

(c) 0.096 kg-m²

(d) 0.128 kg-m²

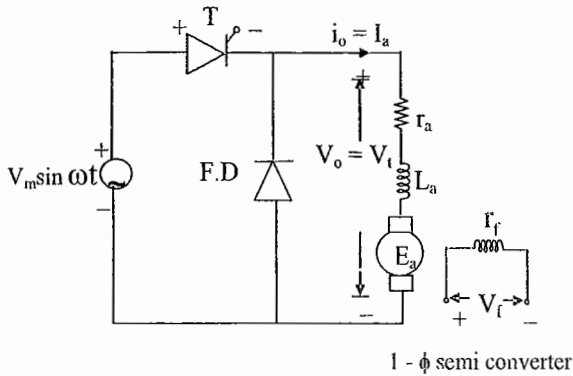


Sol: Generated torque $T_e = J \frac{dw}{dt}$

$$40 = J \left[\frac{\frac{2\pi}{60} [500 - (-1500)]}{0.5} \right]$$

N is in r.p.m and ω is in rad/sec
 $J = 40/418.87 = 0.096 \text{ kg-m}^2$

1 - ϕ Half Wave Rectifier Drive :



V_t = motor terminal voltage, Volts.

I_a = armature current, A.

ϕ = field flux per pole, Wb.

$K_m = K_a \phi$ = torque constant $\frac{N-m}{A}$ or

Emf constant, V-s/rad.

r_a = armature circuit resistance, Ω

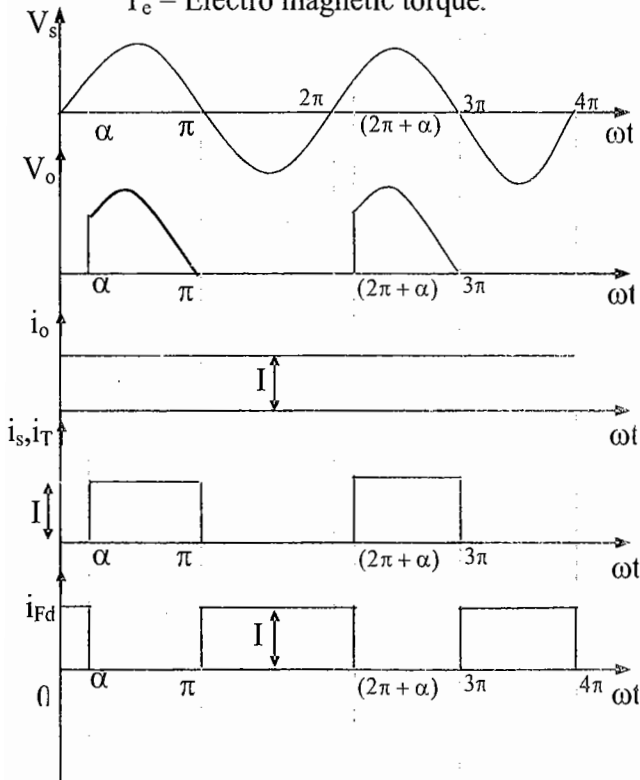
ω_m = angular speed of motor, rad/sec

r_f = field circuit resistance, Ω .

D = Viscous friction constant,

T_L - load torque.

T_e = Electro magnetic torque.



$$V_o = V_t = \frac{V_m}{2\pi} (1 + \cos \alpha_1)$$

$$V_f = \frac{V_m}{2\pi} (1 + \cos \alpha_2)$$

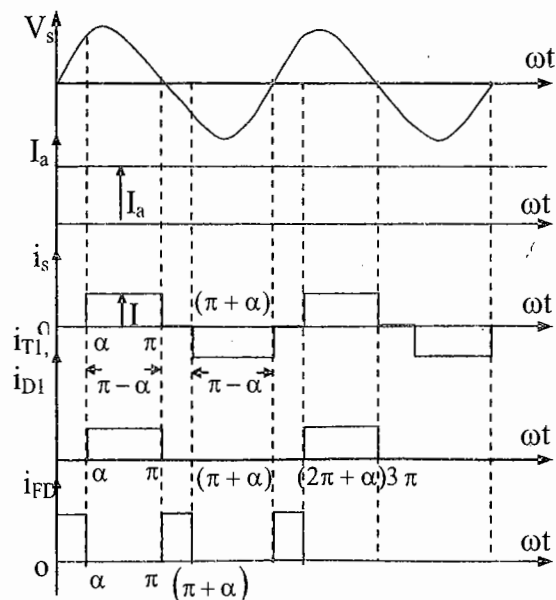
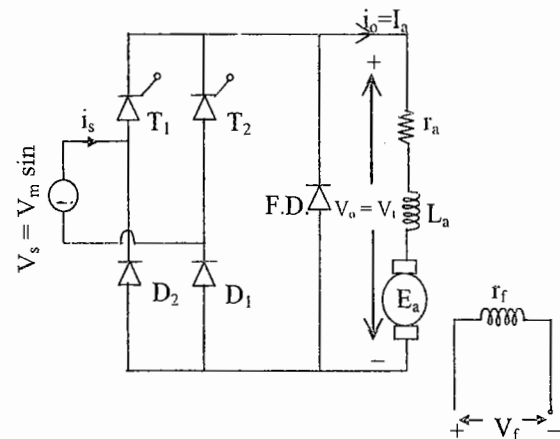
Rms source current

$$I_{sr} = I_{Tr} = I_a \sqrt{\left(\frac{\pi - \alpha_1}{2\pi} \right)}$$

$$\text{Input power factor} = \frac{E_a I_a + I_a^2 r_a}{V_s I_{sr}} = \frac{I_a (E_a + I_a r_a)}{V_s I_{sr}}$$

$$\text{Input power factor} = \frac{V_t \cdot I_a}{V_s \cdot I_{sr}}$$

1- ϕ Semi Converter Drive :





- For a 1 - ϕ semi converter, average output voltage

$$V_o = V_t = \frac{V_m}{\pi}(1 + \cos \alpha)$$

- For field circuit, $V_f = \frac{V_m}{\pi}(1 + \cos \alpha_1)$

- Rms value of source current, $I_{sr} = I_a \sqrt{\frac{\pi - \alpha}{\pi}}$

- Rms value of freewheeling diode current I_{Fd}

$$I_{rms} = I_a \sqrt{\frac{\alpha}{\pi}}$$

- Rms value of thyristor current, $I_{Tr} = I_a$

$$\sqrt{\frac{\pi - \alpha}{2\pi}}$$

$$\begin{aligned} \text{Input power factor} &= \frac{V_t I_a}{V_s I_{sr}} \\ &= \frac{\sqrt{2} V_s (1 + \cos \alpha) I_a \sqrt{\pi}}{\pi V_s I_a \sqrt{\pi - \alpha}} \end{aligned}$$

$$\text{Input pf} = (1 + \cos \alpha) \sqrt{\frac{2}{\pi(\pi - \alpha)}}$$

- A 1 - ϕ semi converter is also called 1 - ϕ half controlled bridge converter.



05. A separately-excited dc motor is fed from two single phase semi converters, one in the armature circuit and the other in the field circuit. Field current is constant at 2A. Motor armature resistance is 0.8Ω and motor constant is $K = 0.5 \text{ V-s/A rad}$. A.C voltage is 230V, 50 Hz for a ripple-free armature current and speed of 1500 r.p.m calculate motor torque for a firing angle of 30° and supply power factor.
- (a) 45.16 Nm, 0.92 lag
(b) 71.56 Nm, 0.46 lag
(c) 71.65 Nm, 0.92 lag
(d) 45.16 Nm, 0.46 lag

Sol: Given data:

$$I_f = 2A, \quad r_a = 0.8 \Omega, \quad K = 0.5 \text{ V-s/rad. A}$$

$$V_s = 230 \text{ V}, \quad N = 1500 \text{ rpm and } \alpha = 30^\circ.$$

$$V_t = E_a + I_a r_a = \frac{V_m}{\pi}(1 + \cos \alpha)$$

$$K_m = K I_f = 0.5 \times 2$$

$$= 1 \text{ V-Sec/rad}$$

$$E_a = K_m \omega_m = 1 \times \frac{2\pi \times 1500}{60} = 157.07 \text{ V}$$

$$157.07 \text{ V} + I_a (0.8) = \frac{\sqrt{2} \times 230}{\pi} (1 + \cos 30^\circ)$$

$$= 193.20$$

$$I_a = 45.16 \text{ A}$$

$$\text{Torque } \tau = k_m \times I_a = 1 \times 45.16 \Rightarrow 45.16 \text{ Nm}$$

$$I_{sr} = I_a \sqrt{\frac{\pi - \alpha}{\pi}} = 45.16 \sqrt{\frac{\pi - 30 \times \frac{\pi}{180}}{\pi}}$$

$$= 41.22 \text{ A}$$

$$\begin{aligned} \text{Power factor} &= \frac{V_t I_a}{V_s I_{sr}} \\ &= \frac{193.20 \times 45.16}{230 \times 41.22} = 0.92 \text{ lag} \end{aligned}$$

Common Data questions (6 & 7)

A separately excited D.C motor fed through 1- ϕ Half controlled bridge converter.

Field current is constant at 2A, motor armature resistance is 0.8Ω and motor constant is $K = 0.5 \text{ V-s/A-rad}$. AC supply voltage is 230V, 50Hz. For a ripple free armature current and speed of 1500 r.p.m firing angle of 30° .

06. Motor current in the armature circuit _____
- (A) 45.12A (B) 90.24A
(C) 22.56A (D) cannot be determined

07. Torque developed by motor _____
- (A) 45.12 Nm (B) 90.24 Nm
(C) 22.56 Nm (D) cannot be determined.

06. Sol: Given data: 1- ϕ semi converter Drive

$$I_f = 2A$$

$$r_a = 0.8 \Omega$$

$$K = 0.5 \text{ V-s/A. rad}$$



$$V_s = 230 \text{ V}$$

$$N = 1500 \text{ r.p.m.}, \alpha = 30^\circ$$

$$V_0 = V_t \Rightarrow V_0 = \frac{V_m}{\pi} [1 + \cos \alpha]$$

$$= \frac{230\sqrt{2}}{\pi} [1 + \cos 30] = 193.17 \text{ V}$$

$$V_t = E_a + I_a r_a = K I_f \omega_m + I_a r_a$$

$$V_0 = 0.5 \times 2 \times 2\pi \times \frac{1500}{60} + I_a \times 0.8 = 193.17$$

$$\Rightarrow I_a = 45.12 \text{ A}$$

07 Sol: Torque = $K I_f I_a$
 $= 0.5 \times 2 \times 45.12 = 45.12 \text{ Nm}$

08. A single – phase half – controlled rectifier is driving a separately excited dc motor. The dc motor has a back emf constant of 0.5 V /rpm. The armature current is 5A without any ripple. The armature resistance is 2Ω . The converter is working from a 230 V, single phase ac source with a firing angle of 30° . Under this operating condition, the speed of the motor will be

(GATE – 04)

- (a) 339 rpm (b) 359 rpm
 (c) 366 rpm (d) 386 rpm

Sol: E.m.f constant $K_b = 0.5 \text{ V/r.p.m}$ (given)

For separately excited d.c motor

$$E_a = K_b N$$

$$\text{and } E = V_0 - I_a R_a$$

$$= \frac{V_m}{\pi} (1 + \cos \alpha) - 5(2)$$

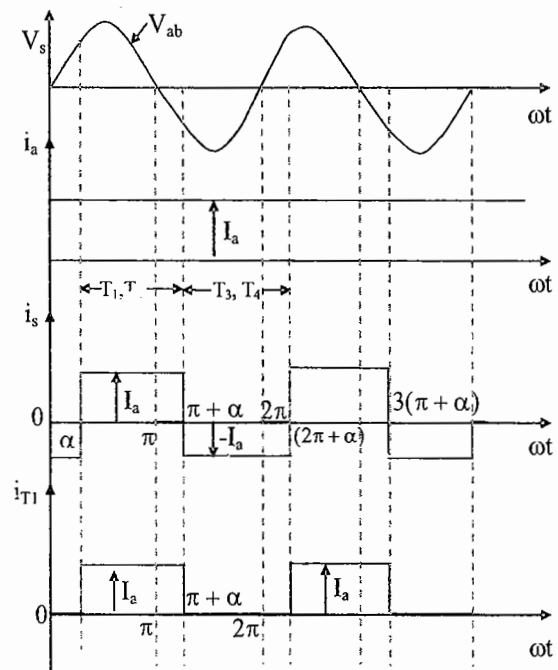
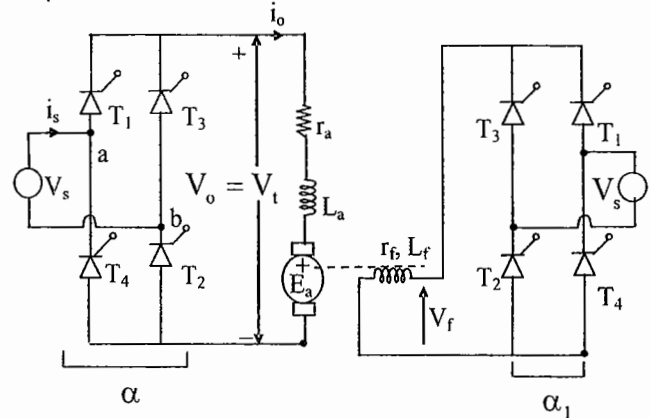
$$= \frac{230 \times \sqrt{2}}{\pi} (1 + \cos 30^\circ) - 10$$

$$E = 183.20$$

$$N = \frac{E}{K_b} = \frac{183.20}{0.5}$$

$$= 366.4 \text{ r.p.m}$$

1- ϕ Full Wave Rectifier Drive:



- For the armature converter 1,

$$V_0 = V_t = \frac{2V_m}{\pi} \cos \alpha \text{ for } 0 < \alpha < \pi$$

- For the field converter 2,

$$V_f = \frac{2V_m}{\pi} \cos \alpha_1 \text{ for } 0 < \alpha_1 < \pi$$

- RMS value of source current,

$$I_{sr} = \sqrt{\left(I_a^2 \cdot \frac{\pi}{\pi} \right)} = I_a$$

- RMS value of thyristor current,

$$I_{Tr} = \sqrt{\left(I_a^2 \cdot \frac{\pi}{2\pi} \right)} = \frac{I_a}{\sqrt{2}}$$



• Input supply p.f. = $\frac{V_t I_a}{V_s I_{sr}} = \frac{2V_m}{\pi} \cos \alpha \frac{I_a \sqrt{2}}{V_m I_a}$

Input p.f. = $\frac{2\sqrt{2}}{\pi} \cos \alpha$

3- ϕ half wave converter drive (or) Rectifier Drive:

$V_o = V_t = \frac{3\sqrt{6}}{2\pi} V_{ph} \cos \alpha$

$I_{sr} = I_{Tr} = I_a \sqrt{\frac{1}{3}}$

3- ϕ Full wave Rectifier Drive:-

$V_o = \frac{3\sqrt{6}}{\pi} V_{ph} \cos \alpha$

$I_{Tr} = I_a \sqrt{\frac{1}{3}}$ and $I_{sr} = I_a \sqrt{\frac{2}{3}}$

3- ϕ Semi Converter Drive:-

For $\alpha_1 < 60^\circ$; $I_{sr} = I_a \sqrt{\frac{2}{3}}$

$V_o = \frac{3\sqrt{6}}{2\pi} V_{ph} (1 + \cos \alpha)$

$I_{Tr} = I_a \sqrt{\frac{1}{3}}$

→ For $\alpha_1 > 60^\circ$, $I_{sr} = I_a \sqrt{\frac{180^\circ - \alpha_1}{180}}$

$I_{sr} = I_a \sqrt{\frac{(180 - \alpha_1)}{360}}$

Chopper drive:- $V_o = \alpha V_s$

Workedout Examples

09. The speed of a 50 kW, 500V, 120A, 1500 r.p.m separately excited dc motor is controlled by a three-phase full converter fed from 400V, 50Hz, supply motor armature resistance is 0.1 Ω . Find the range of firing angle required to obtain speeds between 1000 r.p.m and (-1000 r.p.m) at rated torque ?

- (a) 51.35° , 128.59° (b) 59.92° , 120.07°
(c) 51.35° , 125.41° (d) 59.92° , 117.16°

Sol: Given data

Motor ratings: 50 kW, 500V, 120A, 1500 r.p.m

The motor e.m.f $E_a = V_t - I_a r_a$
 $= 500 - 120(0.1)$
 $= 488V$

For 1000 r.p.m

The motor e.m.f $E_a = \frac{489}{1500} \times 1000$
 $= 325.33V$

$V_t = E_a + I_a r_a$

$\frac{3V_{ml}}{\pi} \cos \alpha = 325.33 + 120(0.1)$

$\cos \alpha = \frac{337 \times \pi}{3 \times \sqrt{2} \times 400}$

$\alpha = 51.35^\circ$

For -1000 r.p.m

$V_t = -325.33 + 120(0.1)$

$\frac{3V_{ml} \cos \alpha}{\pi} = -313$

$\cos \alpha = \frac{-313 \times \pi}{3 \times \sqrt{2} \times 400} \Rightarrow \alpha = 125.41^\circ$

10. A separately excited DC Motor is driven from a 3 phase full converter. The armature current is ripple free. Find 3rd and 5th harmonic components of line currents as a percentage of the fundamental component respectively

- (a) 0% and 20% (b) 0% and -20%
(c) 20% and 0% (d) 0% and -40%

Sol: 3rd harmonic component = 0%

$\frac{5^{th} \text{ harmonic component}}{\text{fundamental component}} = -0.2 = -20\%$

AC DRIVERS

- Induction motors are the one generally used in Ac drives.

Various methods of speed control of induction motors are:

1. Pole changing method.
2. Cascade method of speed control.
3. Stator Voltage Control:

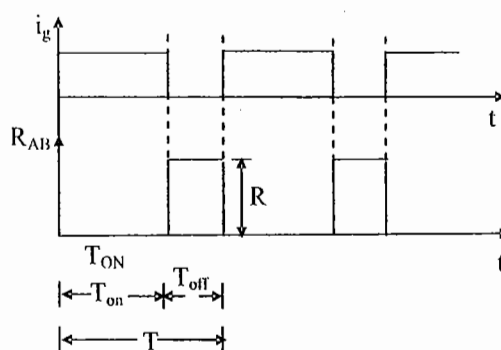
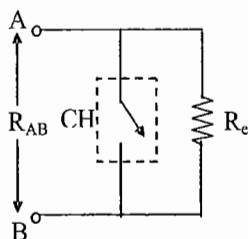
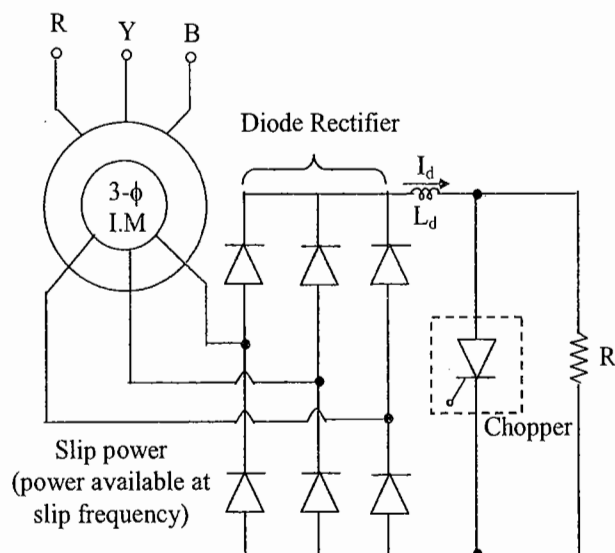
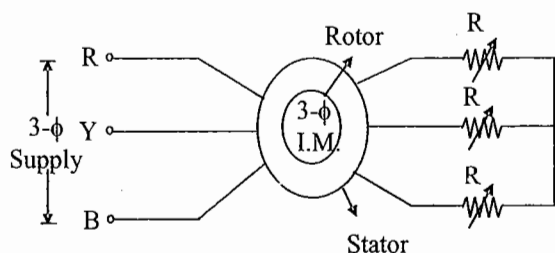
An AC voltage controller will be employed to control the AC voltage applied to a motor.

4. **Stator Frequency Control:**

A cyclo converter will be employed to control the frequency of the AC supply applied to the induction motor.

5. **Stator V/f control:** A cyclo converter will be employed to control the voltage and frequency simultaneously.

Static Rotor Resistance Control:



- When chopper is switched ON, the resistance is zero and when it is switched, OFF, it is 'R'.
- The effective resistance offered to the rotor circuit can be controlled by varying the duty cycle of the chopper.

$$R_{eff} = R \left(\frac{T_{off}}{T} \right)$$

$$= R \frac{(1 - \alpha)T}{T}$$

$$R_{eff} = R(1 - \alpha)$$

- The physical movement of the resistance is being replaced by electronic control of the duty cycle. Hence it is called as 'Static Rotor Resistance Control'.
- In chopper method of speed control, slip power is getting wasted in the external resistance and it leads to poor efficiency of the drive. To recover the slip power, there are two schemes available.

1. **Static Kramer Drive.**

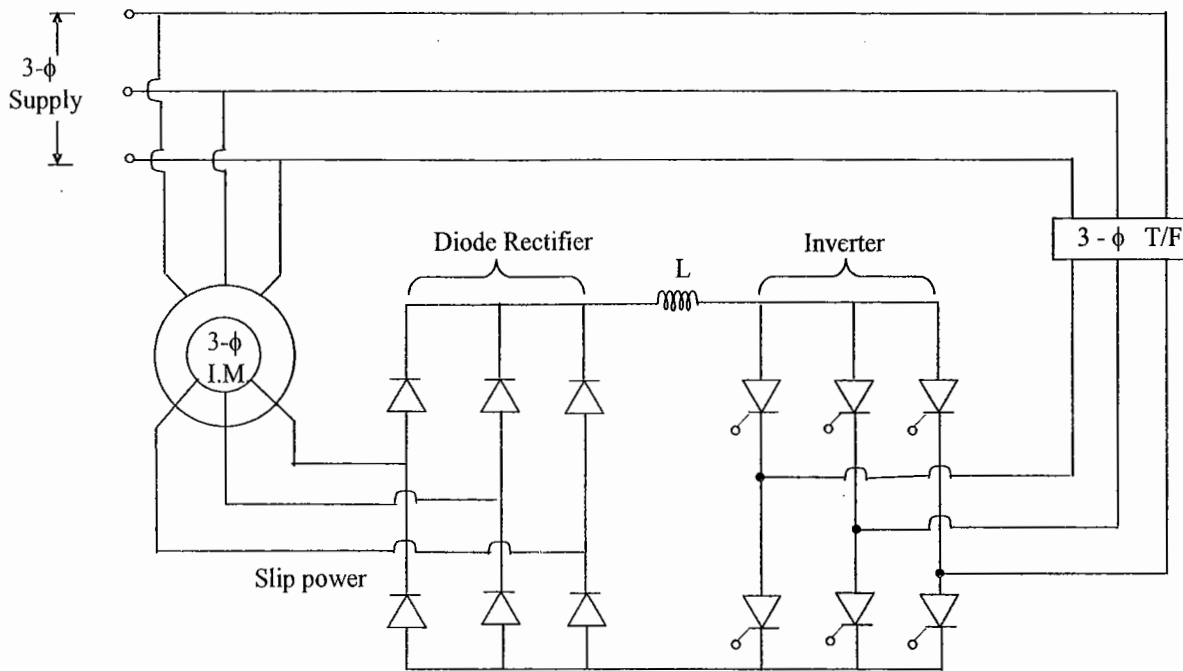
(sub synchronous speed)

2. **Static scherbius Drive.**

(super synchronous speed)

Static Kramer Drive:

- In this method, slip power is getting transferred from rotor circuit to supply lines by adjusting its voltage & frequency values to the line values.
- As the slip power flows from rotor circuit to supply lines it is suitable for only sub synchronous i.e., below the synchronous speed.



Static Scherbius Drive:

There are two configurations to obtain such a drive.

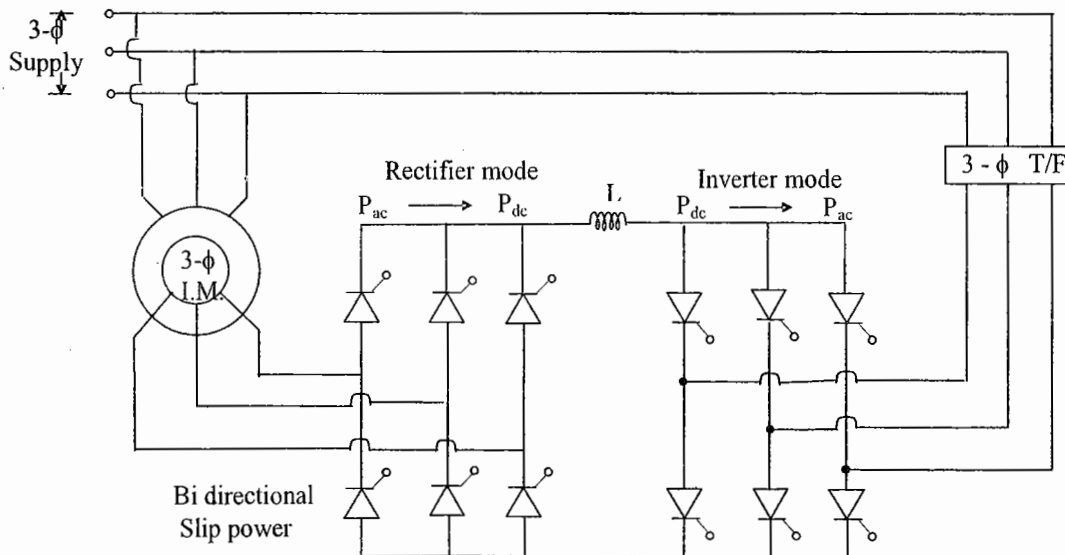
1. DC link scherbius drive
2. Cyclo converter scherbius drive.

1. DC Link Scherbius Drive:

- It consists of two phase controlled bridges, smoothing inductor and transformer.

For sub synchronous speed control Phase Controlled Rectifier- I acts as rectifier ($\alpha < 90^\circ$). P.C. R – II acts as line commutated inverter ($\alpha > 90^\circ$).

- For super synchronous speed control, P.C. R – II acts as rectifier, P. C. R – I will act as a line commutated inverter.





2. Cyclo Converter Scherbius Drive:

- It consists of a cyclo converter, which allows the power flow in both the directions by adjusting its frequency.
- Both the above scherbius drives are suitable for sub – synchronous as well as super synchronous speed.

Workedout Examples

11. A 3 phase wound rotor I.M is controlled by a chopper resistance in its rotor circuit. A resistance of $2\ \Omega$ is connected in the rotor circuit and a resistance of $4\ \Omega$ is additionally connected during off periods of the chopper. The off periods of the chopper is 4 ms. The average resistance in the rotor circuits for the chopper frequency of 200 Hz is

- (a) $\frac{26}{5}\ \Omega$ (b) $\frac{24}{5}\ \Omega$
(c) $\frac{18}{5}\ \Omega$ (d) $\frac{16}{5}\ \Omega$

Sol: $f = 200\text{Hz}$, $T_{\text{off}} = 4\text{ms}$ and $T = \frac{1}{200} = 5\text{msec}$

$$R_{\text{avg}} = \frac{2T_{\text{on}} + (2+4)T_{\text{off}}}{T}$$

$$= \frac{2 \times 1 + 6 \times 4}{5} = \frac{26}{5}\ \Omega$$

Statement for Linked Answer Q.12 & Q.13

A 3-phase 400V, 50Hz, 960 r.p.m, Star connected SRIM has the following per-phase parameters referred to stator.

$$r_1 = 0.1\ \Omega, r_2 = 0.08\ \Omega, x_1 = x_2 = 0.3\ \Omega, X_m = 0.$$

Per – phase turns ratio from rotor to stator = 0.8

Speed of this motor is controlled by a GTO chopper in its rotor circuit. For a speed of 850 r.p.m, the inductor current is 110A and the chopper resistance is $2\ \Omega$.

12. The value of chopper duty cycle is

- (a) 0.656 (b) 0.705
(c) 0.602 (d) 0.672

Sol: Perphase voltage = $\frac{400}{\sqrt{3}} = 230.9\text{V}$

Synchronous speed $N_s = 1000\text{ r.p.m}$

$$N_r = N_s \left[1 - \frac{I_d \cdot R(1-K)}{2.339 \cdot a \cdot V_1} \right]$$

$$850 = 1000 \left[1 - \frac{110 \times 2(1-K)}{2.339 \times 0.8 \times 230.9} \right]$$

$$K = 0.705$$

13. The power loss in chopper is

- (a) 8324W (b) 7139W
(c) 9631.W (d) 7937W

Sol: Power loss in chopper

$$= I_d^2 \times R(1-K)$$

$$= (110)^2 \times 2(1-0.705)$$

$$= 7139\text{W}$$